

Infinitely many quadratic PCF polynomials

A disproof of the first assertion of conjecture 00000007755

earthking11

The first assertion of conjecture 00000007755 says that, for each fixed degree d , there are only finitely many postcritically finite (PCF) polynomials with algebraic coefficients. Here a polynomial is PCF when every critical point is preperiodic under iteration. We disprove this assertion in degree 2.

Let $\overline{\mathbb{Q}}$ be an algebraic closure of $\mathbb{Q}$. For each $a \in \overline{\mathbb{Q}}$, put

$$f_a(X) = (X - a)^2 + a = X^2 - 2aX + (a^2 + a).$$

This polynomial has degree 2 and all its coefficients are algebraic. Its derivative is $f'_a(X) = 2(X - a)$, so its only critical point in $\overline{\mathbb{Q}}$ is a . Moreover $f_a(a) = a$: the critical point is fixed, and hence preperiodic. Therefore f_a is PCF.

The polynomials are pairwise distinct. Indeed, $f_a = f_b$ implies equality of their derivatives. Evaluating at a gives $0 = f'_b(a) = 2(a - b)$, so $a = b$. Because $\overline{\mathbb{Q}}$ is infinite, the set of quadratic PCF polynomials with algebraic coefficients is infinite. This contradicts the first assertion of the conjecture, and thus disproves the conjunction as written. No claim about the separate height asymptotic is necessary.

Formalization. The Lean file defines $K = \text{AlgebraicClosure}(\mathbb{Q})$, $f_a \in K[X]$, preperiodicity as repetition in the evaluation orbit, and PCF using all roots of the formal derivative in K . It proves the degree, critical-point identity, PCF property, and injectivity of $a \mapsto f_a$. The final theorem `TLMC7755.infinite_quadratic_pcf` states that the actual set of degree-2 PCF polynomials in $K[X]$ is infinite. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

Scope. The conjecture quantifies over polynomials, without taking affine-conjugacy classes or imposing a monic-centered normalization. All the examples above are affine conjugate to X^2 ; this argument does not settle a differently worded finiteness assertion modulo affine conjugacy.